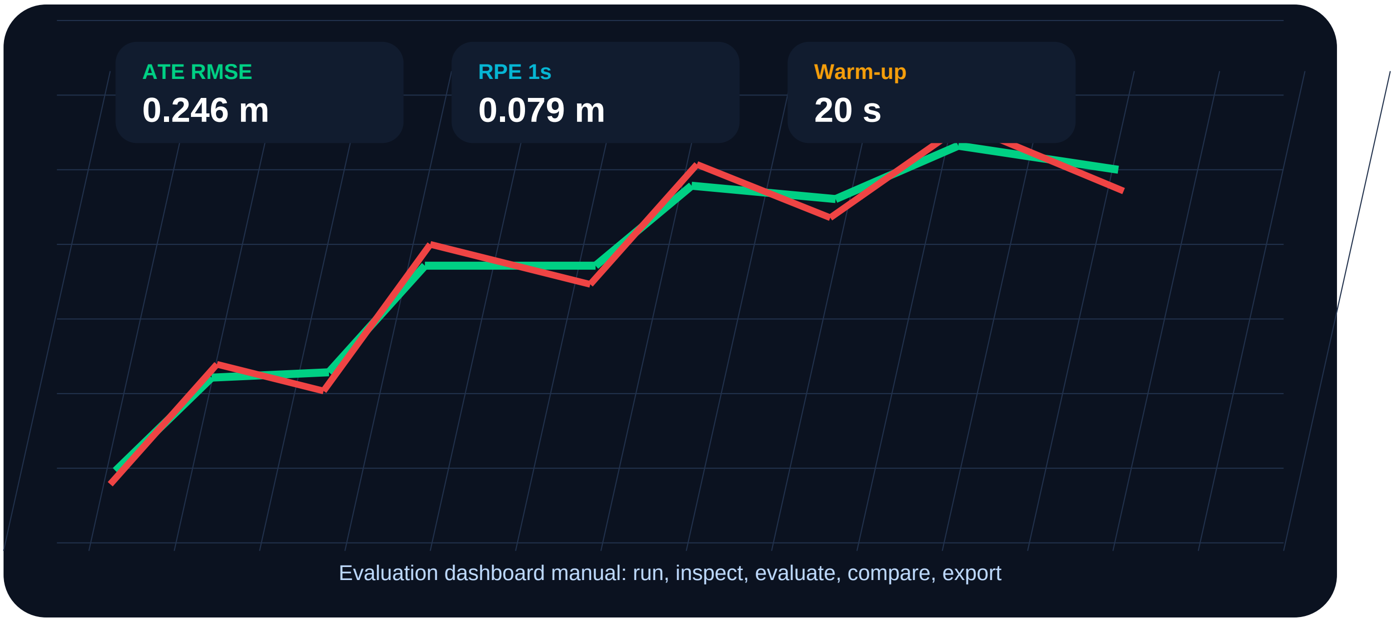


Run VIO experiments, inspect trajectories, evaluate metrics, compare datasets, and export results.

Scope: Phase 9 warm-up-aware dashboard and EuRoC Machine Hall evaluation workflow.

Field	Value
Project	Stereo Visual-Inertial Odometry / MSCKF for UAV relative navigation
Primary GUI	tools/evaluation_dashboard.py
Main dataset family	EuRoC Machine Hall sequences
Main metrics	ATE translation error and translation-only RPE/RTE over 1 second
Warm-up behavior	Optional evaluation-only skip for the initial transient interval

1. Purpose of the Dashboard

The UAV-Airvision Evaluation Dashboard is a graphical tool for running, evaluating, visualizing, and comparing stereo visual-inertial odometry experiments on EuRoC datasets. It is designed to make the VIO workflow understandable without manually opening result files, plotting trajectories, or comparing metrics in separate scripts.

The dashboard is not only a live viewer. It is a full experiment interface. It can launch VIO, display a camera preview, show live and final trajectories, evaluate saved trajectory files against EuRoC ground truth, compute error metrics, export plots, and compare multiple datasets in a global dashboard.

Important concept: the dashboard does not change the estimator math by itself. Evaluation options such as warm-up skip affect metrics and plots only. They do not delete dataset images and they do not change the raw VIO run.

2. Launching the GUI

Open Git Bash in the repository root and activate the project virtual environment. The recommended command for the current warm-up-aware dashboard is:

```
cd "/j/Sattar Run"
source .venv/Scripts/activate

python tools/evaluation_dashboard.py \
  --datasets-root ./datasets \
  --results-root ./results \
  --warmup-skip-seconds 20
```

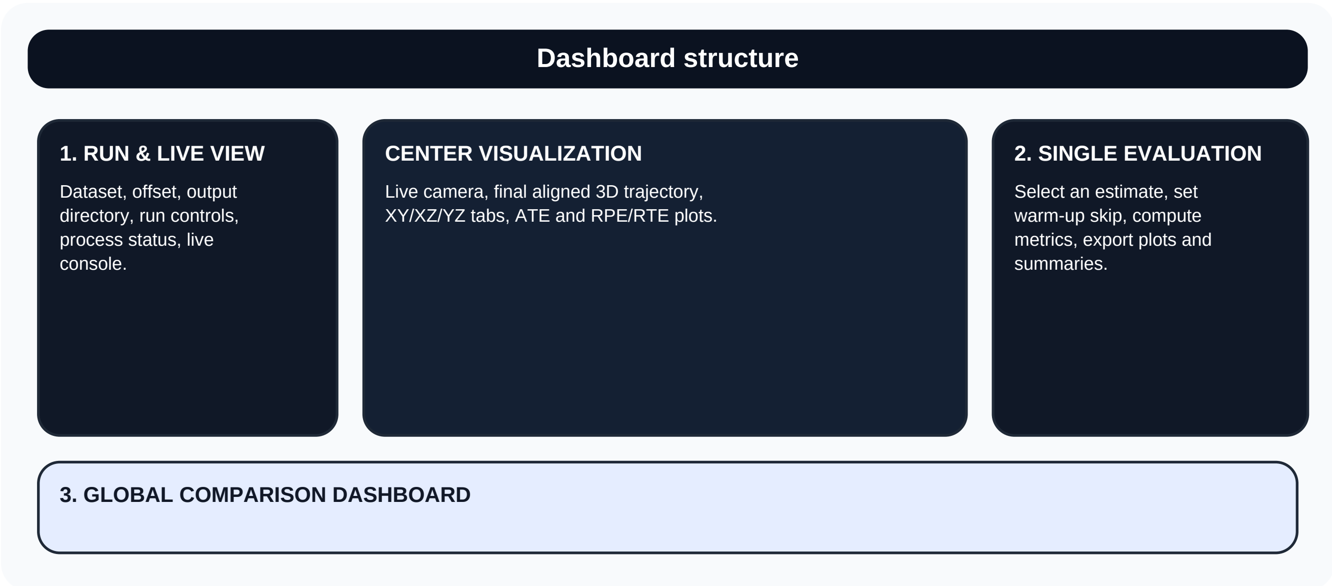
The same dashboard can be launched without warm-up skip by setting the value to zero:

```
python tools/evaluation_dashboard.py \
  --datasets-root ./datasets \
  --results-root ./results \
  --warmup-skip-seconds 0
```

The graphical interface requires optional GUI dependencies. If the dashboard fails with a PyQt, pyqtgraph, PyOpenGL, or matplotlib import error, install the optional GUI stack:

```
python -m pip install PyQt5 pyqtgraph PyOpenGL matplotlib
```

3. Dashboard Layout



The interface is organized into three main working areas. The left sidebar is used to run VIO. The center area is used to visualize camera frames, final aligned trajectories, and metric plots. The right sidebar is used to evaluate one selected result file. The bottom area compares accepted results across datasets.

3.1 Left panel: Run & Live View

The left panel controls experiment execution. It typically contains the dataset selector, offset input, output directory selector, run buttons, stop button, status card, progress bar, and console preview. Use this panel when you want to generate a new VIO output file from a EuRoC dataset.

3.2 Center panel: Visualization area

The center panel contains the main tabs. The Live View tab combines the camera preview, final aligned 3D trajectory, ATE plot, and RPE/RTE plot. Additional tabs provide larger trajectory views in XY, XZ, and YZ planes, plus dedicated error plots.

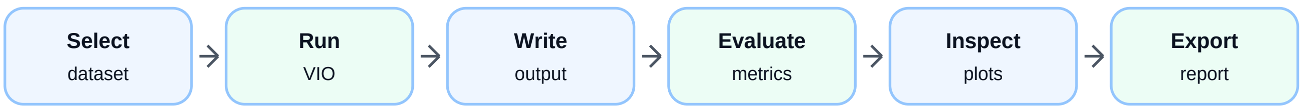
3.3 Right panel: Single Evaluation

The right panel evaluates one selected trajectory output file. It provides an estimate selector, rejected-experiment toggle, warm-up skip input, Evaluate button, metric cards, data-source details, and export buttons. This is the most important panel for checking whether a specific result file is good or bad.

3.4 Bottom panel: Global Comparison Dashboard

The bottom panel summarizes the latest accepted result for each dataset. It includes a comparison table and chart tabs for ATE and RPE/RTE. This area is useful when you want to compare multiple Machine Hall sequences at once.

4. Recommended Workflow



A normal workflow starts by selecting a dataset, choosing an offset and output directory, running VIO, evaluating the generated estimate, inspecting plots, and exporting the results if needed. The dashboard is designed so that the same workflow can be repeated for several datasets without manually editing commands.

4.1 Quick evaluation of an existing result

Use this workflow when the trajectory output already exists under the results directory. Select the result file from the right sidebar, set the warm-up skip value, and press Evaluate. The dashboard recomputes ATE and RPE/RTE from the selected file and updates the plots.

4.2 Running a new VIO experiment

Use this workflow when you want a fresh output file. Select the dataset from the left panel, choose the offset, confirm the output directory, and press Run Selected. The dashboard starts the VIO process and streams console output into the GUI. After the run completes, evaluate the generated file from the right sidebar.

4.3 Comparing datasets

Use Refresh Results to update the bottom comparison dashboard. The table should show one accepted row per dataset by default. Rejected or outlier experiments should remain hidden unless the Show rejected experiments option is enabled.

5. Dataset Selection and Offset

The dataset selector uses the EuRoC Machine Hall registry. Typical dataset keys are MH_01_easy, MH_02_easy, MH_03_medium, MH_04_difficult, and MH_05_difficult. The selected dataset determines the image folder, IMU file, ground-truth file, and expected output filename pattern.

The offset value controls where the VIO run starts inside the dataset. This is different from warm-up skip. Offset changes the actual input data processed by the estimator. Warm-up skip only changes which part of the already-produced estimate is used for evaluation.

Concept	Changes VIO input?	Changes raw output?	Changes metrics?	Typical use
Dataset offset	Yes	Yes	Yes	Start the experiment later in the dataset
Warm-up skip	No	No	Yes	Exclude the unstable startup interval from evaluation
Alignment mode	No	No	Yes	Compare local estimate frame to global ground truth

6. Running VIO From the GUI

When Run Selected is pressed, the dashboard launches the project VIO runner as a child process. The live camera preview and console preview update while the process is running. The generated output file is written under the selected output directory.

The left status card separates run status from evaluation status. This is important because a run may fail while an older evaluated result is still valid, or a run may complete but the new output file may not have been evaluated yet.

Run status	Meaning
Idle	No VIO process is currently running.
Running	The dashboard has launched the VIO process and is receiving output.
Completed	The process ended normally.
Stopped	The user stopped the process.
Run Failed	The process ended with an error or failed to create a valid output.

7. Single Evaluation Panel

The Single Evaluation panel is used to compute metrics for one trajectory estimate file. It is the safest place to verify a result because it shows the selected estimate path, ground-truth path, computation time, alignment mode, acceptance status, and metric cards.

7.1 Result file selector

The result selector lists candidate output files matching the selected dataset. The dashboard should ignore generated analysis artifacts such as cropped warm-up sensitivity files. A valid estimate file normally follows the pattern output_<dataset>_offset<offset>.txt.

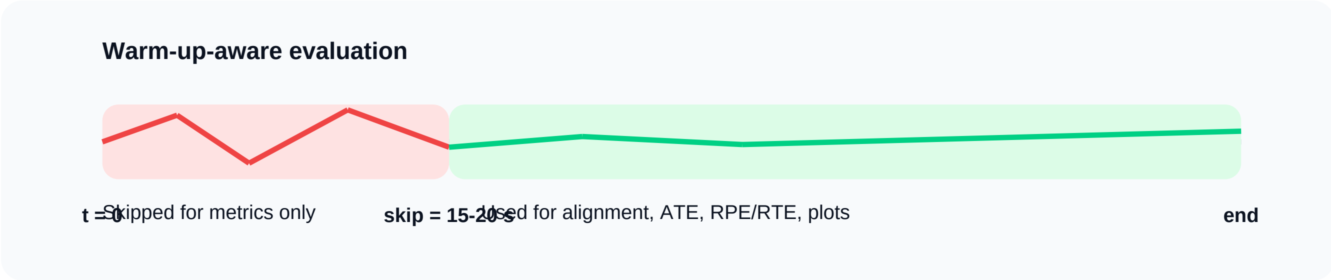
7.2 Show rejected experiments

Rejected experiments are hidden by default. They include known failed or outlier runs such as raw-Jacobian experiments or generated analysis files. Enable this option only when you intentionally want to inspect rejected results.

7.3 Evaluate button

The Evaluate button recomputes metrics from the selected estimate and ground truth. It refreshes metric cards, final trajectory plots, error plots, and the data-source card. Evaluation should be repeated after changing the warm-up skip value.

8. Warm-Up Skip



Warm-up skip is an evaluation-only option. It removes the first N seconds of the estimate before alignment and metric computation. This is useful because the current VIO implementation often has an initial transient during the first 10-20 seconds after startup.

Warm-up skip does not fix the estimator. It only allows the dashboard to report post-startup performance separately from startup-inclusive performance. The raw output file is preserved. The dataset is not modified.

A value of 0 seconds means the full estimate is evaluated from the first output sample. A value of 15 or 20 seconds means the initial interval is excluded from ATE/RPE computation and from the final aligned plot.

Warm-up value	Interpretation	Recommended use
0 s	Full startup-inclusive metric.	Use when debugging the true initial transient.
10 s	Moderate startup exclusion.	Use as a quick sensitivity check.
15 s	Common post-startup evaluation point.	Use when comparing MH_05 phase results.
20 s	Stronger post-startup evaluation point.	Use for stable-performance dashboard review.

9. Understanding Alignment and Origin

The final trajectory view displays an aligned estimate and interpolated ground truth in a shared frame. This is different from the raw VIO coordinate origin. The raw VIO estimator may start near its own local origin, while EuRoC ground truth is stored in a global reference frame. SE(3) alignment estimates a rigid transform that places the estimate into the ground-truth frame for fair position-error comparison.

The 3D origin in the dashboard is a display reference. The selected estimate point does not have to lie exactly on the origin. The important quantity is the error between the estimate and the ground truth at the same timestamp.

When warm-up skip is enabled, the first visible sample is no longer the first raw VIO output sample. It is the first sample after the skipped interval. Therefore, the visible trajectory may appear to start from a different location than the raw VIO start. This is expected and is not automatically an algorithmic bug.

Object	What it represents
Red trajectory	Aligned estimated trajectory from the selected VIO output file.
Green trajectory	Interpolated EuRoC ground-truth trajectory in the same aligned frame.
Selected marker	The sample currently selected by the trajectory inspector slider.
Start after warm-up	The first estimate sample used after applying warm-up skip.
Error value	Distance between estimate and ground truth at the selected sample.

10. Metrics

Metric card interpretation

The dashboard currently focuses on translation metrics. These metrics are useful for comparing position accuracy, but they do not yet fully describe orientation error or full SE(3) pose error.

Metric	Meaning	Lower is better?	Notes
ATE RMSE	Root-mean-square absolute translation error after alignment.	Yes	Primary position accuracy metric.
ATE mean	Average absolute translation error.	Yes	Less sensitive to single spikes than max error.
ATE median	Middle error value.	Yes	Useful for understanding typical performance.
ATE max	Worst absolute translation error.	Yes	Highlights outliers and spikes.
RPE/RTE 1s RMSE	Translation error over 1-second relative motion intervals.	Yes	Current implementation is translation-only.
Aligned samples	Number of estimate samples used after filtering and alignment.	Higher is not always better	Can decrease when warm-up skip is enabled.
Overlap duration	Time interval shared by estimate and ground truth.	Context-dependent	Shows how much of the sequence was evaluated.

11. Visualization Tabs

11.1 Live View tab

The Live View tab combines the camera preview, the final aligned 3D trajectory, the ATE plot, and the RPE/RTE plot. It is the most useful tab for a quick overall check.

11.2 Trajectory XY, XZ, and YZ tabs

These tabs show 2D projections of the aligned trajectory. XY is usually the most intuitive top-down view. XZ and YZ help detect vertical drift and altitude inconsistency.

11.3 ATE Plot tab

The ATE plot shows absolute translation error over time. Large early values usually indicate startup transient behavior. Late rising values may indicate drift, bad feature updates, or accumulated IMU error.

11.4 RPE/RTE Plot tab

The RPE/RTE plot shows relative translation error over fixed intervals. It is useful for detecting local motion inconsistency even when global alignment makes the trajectory look acceptable.

11.5 3D trajectory inspector

The 3D trajectory inspector includes Prev, Next, Latest, and slider controls. The text under the plot reports the selected sample index, relative time, selected estimate coordinates, and error at that sample.

12. Reading Common Visual Patterns

Pattern	Likely interpretation	What to check
High error only at the beginning	Initial transient or initialization issue.	Compare skip 0, 15, and 20 seconds.
Trajectory shape matches but is shifted	Alignment/origin interpretation issue or local-frame output.	Use SE(3) alignment and inspect selected-sample error.
RPE spikes but ATE looks moderate	Local motion inconsistency.	Check update logs and feature geometry around spike timestamps.
ATE grows steadily over time	Long-term drift.	Inspect IMU propagation, bias estimates, and feature tracking quality.
Very large outlier metrics	Failed experiment or corrupted output.	Enable rejected experiments only intentionally.

13. Global Comparison Dashboard

The bottom dashboard summarizes accepted results for multiple datasets. It is intended for high-level comparison rather than detailed debugging of one run. The table should normally show one row per dataset. The chart tabs make it easier to compare ATE RMSE and RPE/RTE 1s RMSE across datasets.

If rejected experiments are hidden, the dashboard avoids extreme outliers that would otherwise distort chart scales. If rejected experiments are shown, the charts may use a different scale or appear visually compressed because failed runs can have very large errors.

Column	Description
Dataset	EuRoC sequence key.
Status	Accepted, warning, rejected, missing, or failed.
Output	Shortened path to the selected estimate file.
ATE RMSE	Main absolute position error.
RPE 1s	Translation-only relative error over one second.
ATE Mean	Average absolute translation error.
Aligned	Number of aligned samples.
Overlap	Evaluated overlap duration.

14. Exports

The dashboard can export plots and summary files. Exported files are generated artifacts and should not be committed unless explicitly approved as documentation assets.

Action	Purpose
Export Current Plot	Save the currently displayed plot for reports or presentations.
Export Summary	Save the global metric summary table.
Open Results Folder	Open the directory containing generated outputs and evaluation artifacts.
Refresh Results	Re-scan result files and recompute or reload metric rows.

15. Data Hygiene Rules

Do not commit raw result directories, runtime logs, generated CSV files, generated PNG plots, cropped warm-up estimates, or experiment bundles unless the file is explicitly intended as a documentation artifact.

The dashboard must distinguish raw VIO outputs from derived analysis outputs. Cropped files created for warm-up sensitivity analysis should not be selected as raw estimates. Otherwise, the dashboard may apply warm-up skip twice and remove too many samples.

```
Do not commit:
results/
*.csv generated by evaluation
*.png generated by plotting
runtime logs
results/warmup_sensitivity_existing_outputs/
cropped_estimates/
phase review bundles
```

16. Validation Commands

Before trusting a dashboard branch, run the nonvisual validation commands. They check that the GUI can be constructed and that data-flow evaluation works without opening the full interactive interface.

```
python -m py_compile tools/evaluation_dashboard.py tools/evaluate_trajectory.py tools/evaluate_euroc_batch.py

python tools/evaluation_dashboard.py \
  --ui-self-check \
  --datasets-root ./datasets \
  --results-root ./results \
  --warmup-skip-seconds 20

python tools/evaluation_dashboard.py \
  --validate-data-flow \
  --datasets-root ./datasets \
  --results-root ./results \
  --warmup-skip-seconds 20
```

17. Troubleshooting

Problem	Likely cause	Fix
GUI does not start	Missing optional GUI dependency.	Install PyQt5, pyqtgraph, PyOpenGL, and matplotlib.
Camera preview shows FPS 0.0	No active run or preview is showing a static first frame.	Run VIO or confirm live frame updates are enabled.
Evaluation fails with too few samples	Warm-up skip is larger than the available estimate duration, or a cropped file was selected.	Use a smaller skip and avoid cropped_estimates files.
Selected file is from warmup_sensitivity_existing_outputs	Dashboard candidate filtering missed generated artifacts.	Filter generated/cropped outputs from candidate discovery.
ATE looks worse with skip 0	Initial transient is included.	Compare with skip 15 or 20 seconds for post-startup performance.
3D point does not start at origin	The plot is aligned to ground truth and uses a shared display frame.	Inspect the selected-sample error; this is not automatically a bug.
Charts look compressed	Rejected/outlier experiments are visible.	Disable Show rejected experiments for normal comparison.

18. Recommended Use Cases

18.1 Quick health check

Launch the dashboard with warm-up skip set to 20 seconds, evaluate the latest accepted result, and check ATE RMSE, RPE 1s RMSE, and the final aligned 3D trajectory. This gives a stable-performance overview.

18.2 Startup transient debugging

Set warm-up skip to 0 seconds and inspect early ATE spikes. Compare the same result with skip 15 and skip 20. If metrics improve dramatically after skip, the issue is likely concentrated in the initialization or first visual updates.

18.3 Experiment comparison

Use the bottom global dashboard to compare accepted results across datasets. Keep rejected experiments hidden unless the purpose is to inspect failures.

18.4 Report preparation

Evaluate the selected output, export the current plot, export the summary table, and include the warm-up skip value in any written report. Never report warm-up-aware metrics without stating the skip interval.

19. Current Limitations

The dashboard currently reports translation-focused metrics. It does not yet provide full orientation error, full SE(3) RPE, or covariance consistency metrics. The live preview depends on disk speed, image size, Qt/OpenGL performance, and how often the VIO process emits progress information.

Warm-up-aware evaluation is a measurement tool, not a final algorithmic fix. The next technical phase should investigate the first 10-20 seconds of VIO startup to identify whether the root cause is IMU initialization, feature geometry, covariance confidence, timestamp ordering, or early MSCKF update behavior.

20. Glossary

Term	Meaning
VIO	Visual-inertial odometry, estimating motion from cameras and IMU.
MSCKF	Multi-State Constraint Kalman Filter, the backend estimator style used in this project.
EuRoC	A benchmark dataset family containing synchronized stereo images, IMU, and ground truth.
ATE	Absolute Trajectory Error after alignment.
RPE/RTE	Relative Pose/Translation Error over a fixed interval; currently translation-only in this project.
SE(3) alignment	Rigid alignment of estimated positions to ground truth without scale change.
Warm-up skip	Evaluation-only exclusion of the first N seconds of the estimate.
Rejected experiment	Known failed, outlier, generated, or non-final output excluded from default comparison.

21. Final Notes

Use the dashboard as the main interface for controlled VIO evaluation. Always know whether you are looking at startup-inclusive metrics or warm-up-aware metrics. Treat warm-up skip as a diagnostic and reporting tool, not as proof that the estimator is fully fixed. For algorithmic debugging, focus on the first 20 seconds and use the dashboard to confirm whether each fix improves both raw and post-warm-up performance.